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MOTOR-DRIVEN COMPRESSOR

Abstract

An inverter of a motor-driven compressor includes a circuit board on which an electronic component is mounted. The housing includes a motor housing member that defines a suction chamber, an inverter housing member that defines an inverter accommodating chamber, and a partition wall that separates the suction chamber and the inverter accommodating chamber from each other. The electronic component includes a main body that acts to remove noise. The circuit board faces the partition wall. The main body faces an outer peripheral surface of the motor housing member in a radial direction of the rotary shaft. A projected area of the main body onto the circuit board in the axial direction is smaller than a projected area of the main body onto the outer peripheral surface of the motor housing member in the radial direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-028331, filed on Feb. 28, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to a motor-driven compressor.

2. Description of Related Art

[0003] For example, International Publication No. 2020/090701 discloses a motor-driven compressor that includes a compression mechanism, an electric motor, an inverter controller, and a housing. The housing includes a compressor housing member that accommodates the compression mechanism, a motor housing member that accommodates the electric motor, an inverter housing member that accommodates the inverter controller, and a cover connected to the inverter housing member.

[0004] The inverter housing member includes a tubular side wall and a partition wall that closes the rear side of the side wall. The cover closes a space defined by the side wall and the partition wall. An inverter chamber, which accommodates the inverter controller, is defined in the space defined by the side wall, the partition wall, and the cover.

[0005] In the motor-driven compressor disclosed in the above publication, each of the inverter housing member and the cover includes a projecting portion that projects radially outward from the outer peripheral surface of the motor housing member, in a side view of the motor housing member as seen in the axial direction. These projecting portions expand the inverter chamber.

[0006] In the motor-driven compressor disclosed in the above publication, a portion including the projecting portion of the inverter housing member and the projecting portion of the cover may vibrate. Such vibration of the projecting portion of the inverter housing member and the projecting portion of the cover may generate noise.

SUMMARY

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0008] In one general aspect, a motor-driven compressor includes a rotary shaft, a motor configured to rotate the rotary shaft, a compression unit configured to compress a fluid by rotation of the rotary shaft, an inverter configured to drive the motor, and a housing that accommodates the rotary shaft, the compression unit, the motor, and the inverter. The inverter includes a circuit board on which a drive element configured to drive the motor and an electronic component configured to remove noise are mounted. The housing includes a motor housing member that accommodates the motor and defines a suction chamber configured to draw in the fluid, a compression unit housing member that accommodates the compression unit and is configured to discharge the compressed fluid, an inverter housing member that defines an inverter accommodation chamber that accommodates the inverter, and a partition wall that separates the suction chamber and the inverter accommodating chamber from each other. The electronic component includes a main body that acts to remove noise, and a connection portion that electrically connects the main body and the circuit

board to each other. The inverter accommodation chamber includes a first accommodation space in which the circuit board is disposed to face the partition wall, and a second accommodation space that opens in an axial direction of the rotary shaft and is arranged such that the main body faces an outer peripheral surface of the motor housing member in a radial direction of the rotary shaft. A projected area of the main body onto the circuit board in the axial direction is smaller than a projected area of the main body onto the outer peripheral surface of the motor housing member in the radial direction.

[0009] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a partially cut away diagram showing a motor-driven compressor according to an embodiment.

[0011] FIG. 2 is a cross-sectional view illustrating part of the motor-driven compressor shown in FIG. 1.

[0012] FIG. 3 is a perspective view of the housing body and the inverter shown in FIG. 2.

[0013] FIG. 4 is a perspective view of the magnetic component shown in FIG. 3.

[0014] FIG. 5 is an explanatory cross-sectional view illustrating operation of the motor-driven compressor shown in FIG. 1.

[0015] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0016] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0017] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0018] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0019] A motor-driven compressor **10** according to an embodiment will now be described with reference to FIGS. 1 to 5. The motor-driven compressor **10** of the present embodiment is used in a vehicle air conditioner.

[0020] As shown in FIG. 1, the motor-driven compressor **10** includes a housing **11**, a rotary shaft **13**, a compression unit **14**, a motor **15**, and an inverter **16**. The housing **11** accommodates the rotary shaft **13**, the compression unit **14**, the motor **15**, and the inverter **16**. The motor **15** rotates the rotary shaft **13**. The compression unit **14** is driven by rotation of the rotary shaft **13** to compress a refrigerant, which is a fluid. The inverter **16** drives the motor **15**.

Housing

[0021] The housing **11** includes a housing body **20**, a first cover **21**, and a second cover **22**. Each of the housing body **20**, the first cover **21**, and the second cover **22** is made of metal. For example, each of the housing body **20**, the first cover **21**, and the second cover **22** is made of aluminum.

[0022] As shown in FIGS. 2 and 3, the housing body **20** includes a plate-shaped partition wall **23**, a

tubular motor housing member **24**, and a tubular inverter housing member **25**. The motor housing member **24** extends in a tubular shape from an outer peripheral portion of the partition wall **23** in the thickness direction of the partition wall **23**. The inverter housing member **25** extends from a part of the outer peripheral portion of the partition wall **23** in a direction opposite to the direction in which the motor housing member **24** extends.

[0023] As shown in FIG. **1**, the direction in which a central axis L of the rotary shaft **13** extends is defined as an axial direction X. The axial direction of the motor housing member **24** agrees with the axial direction X of the rotary shaft **13** and also agrees with the thickness direction of the partition wall **23**. Accordingly, the tubular motor housing member **24** extends in the axial direction X of the rotary shaft **13**.

[0024] As shown in FIG. **3**, the motor housing member **24** includes a passage forming portion **12**. The passage forming portion **12** protrudes from an outer peripheral surface **24a** of the motor housing member **24**. The passage forming portion **12** is disposed between a central portion of the motor housing member **24** and the partition wall **23** in the axial direction of the motor housing member **24**. A suction passage **12a** is defined in the passage forming portion **12**. The suction passage **12a** is connected to a first end of an external refrigerant circuit (not shown). As shown in FIG. **1**, the motor housing member **24** and the partition wall **23** define a suction chamber S1.

[0025] As shown in FIG. **3**, the outer shape of the inverter housing member **25** is larger than the outer shape of the motor housing member **24**. The inverter housing member **25** includes a base **27**, which is continuous with the outer edge of the partition wall **23**, and an extension **28**, which is not continuous with the partition wall **23**. The extension **28** is located outward of an outer peripheral surface **24a** of the motor housing member **24** in a first direction Y, which is orthogonal to the axial direction X. The first direction Y agrees with one of radial directions R of the rotary shaft **13**. In a side view of the motor housing member **24** as seen in the axial direction X, the extension **28** protrudes outward from the outer peripheral surface **24a** of the motor housing member **24**. In other words, the extension **28** protrudes outward beyond the outer peripheral surface **24a** of the motor housing member **24** in a radial direction R of the rotary shaft **13**.

[0026] As shown in FIG. **2**, unlike the base **27**, the extension **28** extends in the axial direction X from the partition wall **23** on the same side as the motor housing member **24**. Accordingly, the dimension of the extension **28** in the axial direction X is larger than the dimension of the base **27** in the axial direction X. The extension **28** includes a first end **28a** and a second end **28b**. Each of the first end **28a** and the second end **28b** is an end of the extension **28** in the axial direction X. The first end **28a** of the extension **28** is located at the same position as a distal end **27a** of the base **27** in the axial direction X. The second end **28b** of the extension **28** is located on the opposite side of the partition wall **23** from the first end **28a** in the axial direction X. A portion of the extension **28** that extends from the partition wall **23** on the same side as the motor housing member **24** in the axial direction X is located on the outer periphery of the motor housing member **24**.

[0027] The extension **28** includes a closing wall **26** that connects the outer peripheral surface **24a** of the motor housing member **24** to the second end **28b** of the inverter housing member **25**. The closing wall **26** is located on the same side in the axial direction X of the partition wall **23** as the motor housing member **24**. Specifically, the closing wall **26** is connected to the outer peripheral surface **24a** of the motor housing member **24** at a position that is on the same side of the partition wall **23** as the motor housing member **24** in the axial direction X and away from the partition wall **23**. Accordingly, a space S is defined inside the extension **28** by the outer peripheral surface **24a** of the motor housing member **24**, the inner peripheral surface of the extension **28**, and the inner surface of the closing wall **26**. The space S is defined so as to be recessed in relation to the partition wall **23** in the same direction in which the motor housing member **24** extends. Therefore, the space S is open in the axial direction X of the rotary shaft **13**.

[0028] A connector **17** is provided on the closing wall **26**. The connector **17** extends in the axial direction X from the closing wall **26** on the side opposite to the inverter housing member **25**. The

connector **17** is electrically connected to a power supply (not shown) mounted on a vehicle (not shown). The connector **17** is electrically connected to a circuit board **51**, which will be discussed below, by a wire **17a**.

[0029] As shown in FIG. **1**, the partition wall **23** includes a boss **29**. The boss **29** extends in the same direction as the direction in which the motor housing member **24** extends from the partition wall **23**. The boss **29** includes a recess **29a**. The recess **29a** is recessed from a central portion of the distal end face of the boss **29**.

[0030] The outer shape of the first cover **21** corresponds to the outer shape of the inverter housing member **25**. As shown in FIG. **2**, the first cover **21** is coupled to the inverter housing member **25**. Specifically, the first cover **21** is coupled to the distal end **27a** of the base **27** and the first end **28a** of the extension **28**. The first cover **21** closes the opening of the inverter housing member **25**. The first cover **21** also closes an opening formed in the extension **28** of the inverter housing member **25**. The extension **28** and a portion of the first cover **21** that closes the opening formed in the extension **28** of the inverter housing member **25** form a projecting wall **111**. As described above, in a side view of the motor housing member **24** as seen in the axial direction X, the extension **28** protrudes outward from the outer peripheral surface **24a** of the motor housing member **24**. Therefore, the housing **11**, in a side view of the motor housing member **24** as seen in the axial direction X, includes the projecting wall **111**, which extends outward beyond the outer peripheral surface **24a** of the motor housing member **24**. Hereinafter, in the side view of the motor housing member **24** as seen in the axial direction X, the direction in which the projecting wall **111** extends outward beyond the outer peripheral surface **24a** of the motor housing member **24** is defined as the first direction Y. The direction orthogonal to both the axial direction X and the first direction Y is defined as a second direction Z.

[0031] The partition wall **23**, the extension **28**, a portion of the outer peripheral surface **24a** of the motor housing member **24**, the inner peripheral surface of the inverter housing member **25**, and the inner surface of the first cover **21** define an inverter accommodating chamber S2 that accommodates the inverter **16**. Accordingly, the inverter housing member **25** defines the inverter accommodating chamber S2. The inverter accommodating chamber S2 is also defined inside the projecting wall **111**. The partition wall **23** separates the suction chamber S1 from the inverter accommodating chamber S2 in the axial direction X.

[0032] The inverter accommodating chamber S2 includes a first accommodation space S21, which faces the partition wall **23** in the axial direction X, and a second accommodation space S22, which is located outward of the outer peripheral surface **24a** of the motor housing member **24** in a side view of the motor housing member **24** as seen in the axial direction X. The second accommodation space S22 is formed by closing the space S defined inside the projecting wall **111** with the first cover **21**. Therefore, the second accommodation space S22 is an accommodation space defined inside the projecting wall **111**.

[0033] The second accommodation space S22 is located alongside the suction chamber S1 in the first direction Y with the motor housing member **24** interposed therebetween. Therefore, the second accommodation space S22 is disposed outward of the outer peripheral surface **24a** of the motor housing member **24** in the radial direction R of the rotary shaft **13**. In the following description, a section of the outer peripheral surface **24a** of the motor housing member **24** that defines the second accommodation space S22 is referred to as a defining surface **240**.

[0034] As shown in FIG. **1**, the second cover **22** is coupled to a distal end of the motor housing member **24**. The second cover **22** closes the opening of the motor housing member **24**. The second cover **22** includes a discharge port **22a**. The discharge port **22a** is connected to a second end of the external refrigerant circuit (not shown) opposite to the first end, to which the suction passage **12a** is connected.

[0035] The motor-driven compressor **10** is mounted on the vehicle in such an orientation that the suction chamber S1 and the first accommodation space S21 of the inverter accommodating

chamber **S2** are alongside each other in the horizontal direction. However, the mounting orientation of the motor-driven compressor **10** with respect to the vehicle may be changed as necessary. The first accommodation space **S21** of the inverter accommodating chamber **S2** is located above or below the second accommodation space **S22** in the vertical direction.

[0036] The rotary shaft **13** is accommodated in the motor housing member **24** of the housing **11**. The rotary shaft **13** extends in the axial direction of the motor housing member **24**. A first shaft end of the rotary shaft **13** is inserted into the recess **29a** of the boss **29**. The first shaft end of the rotary shaft **13** is rotatably supported by the boss **29** with a bearing **18a**. A second shaft end of the rotary shaft **13**, which is located at the side opposite to the first shaft end, is rotatably supported by a shaft support member (not shown).

[0037] The compression unit **14** is accommodated in the motor housing member **24** of the housing **11**. The compression unit **14** is disposed between the motor **15** and the second cover **22** in the axial direction **X**. Thus, the motor housing member **24** includes a compression unit housing member that accommodates the compression unit **14**.

[0038] The motor **15** is accommodated in the suction chamber **S1**. That is, the suction chamber **S1** also acts as a motor accommodating chamber that accommodates the motor **15**. Therefore, the motor housing member **24** accommodates the motor **15** and defines the suction chamber **S1**, into which a fluid is drawn. The motor **15** includes a rotor **41** and a stator **42**. The rotor **41** includes a cylindrical rotor core **41a** and permanent magnets **41b**. The rotor core **41a** is fixed to the rotary shaft **13**. The permanent magnets **41b** are embedded in the rotor core **41a**. The permanent magnets **41b** are disposed at equal intervals in the circumferential direction of the rotor core **41a**. The stator **42** surrounds the rotor **41**. The stator **42** has a cylindrical stator core **42a** and a motor coil **42b**. The stator core **42a** is fixed to the inner peripheral surface of the motor housing member **24**. The motor coil **42b** is wound around the stator core **42a**.

[0039] When the motor coil **42b** is energized, the rotor **41** rotates. The rotary shaft **13** rotates integrally with the rotor **41**. When the rotary shaft **13** operates, the compression unit **14** is driven. When the compression unit **14** is driven, refrigerant is drawn into the suction chamber **S1** from the external refrigerant circuit through the suction passage **12a**. The refrigerant drawn into the suction chamber **S1** is compressed by the compression unit **14**. The refrigerant compressed by the compression unit **14** is discharged to a discharge chamber (not shown) defined in the motor housing member **24**, and is then discharged through the discharge port **22a** to the external refrigerant circuit. The compression unit **14** may be a scroll type, a piston type, or a vane type. As described above, the housing **11** includes the motor housing member **24**, which accommodates the motor **15** and defines the suction chamber **S1** into which the refrigerant is drawn, the compression unit housing member, which accommodates the compression unit **14** and discharges compressed fluid, the inverter housing member **25**, which defines the inverter accommodating chamber **S2** that accommodates the inverter **16**, and the partition wall **23**, which separates the suction chamber **S1** from the inverter accommodating chamber **S2**.

Inverter

[0040] As shown in FIGS. **2** and **3**, the inverter **16** accommodated in the inverter accommodating chamber **S2** includes the circuit board **51**, on which an inverter circuit **52** including drive elements **52a** for driving the motor **15**, a magnetic component **53**, and two capacitors **54** are mounted. The magnetic component **53** includes a choke coil **55** and a conductive ring **56**. The magnetic component **53** and the capacitors **54** are electronic components that remove noise.

[0041] The thickness direction of the circuit board **51** agrees with the axial direction **X**. The circuit board **51** has a first mounting surface **511** and a second mounting surface **512** located on a side opposite to the first mounting surface **511** in the thickness direction. The outer shape of the circuit board **51** is larger than the outer shape of the motor housing member **24**. The circuit board **51** includes a first board portion **51a**, which is accommodated in the first accommodation space **S21** of the inverter accommodating chamber **S2**, and a second board portion **51b**, which is accommodated

in the second accommodation space S22 of the inverter accommodating chamber S2. The first board portion **51a** of the circuit board **51** is disposed in the first accommodation space S21 of the inverter accommodating chamber S2 so as to face the partition wall **23**. Therefore, the inverter accommodating chamber S2 includes the first accommodation space S21, in which the circuit board **51** is disposed so as to face the partition wall **23**.

[0042] The second accommodation space S22, in which the second board portion **51b** of the circuit board **51** is accommodated, is open in the axial direction X of the rotary shaft **13**. The second board portion **51b**, which is a part of the circuit board **51**, is accommodated in the second accommodation space S22 of the inverter accommodating chamber S2 such that the first mounting surface **511** is orthogonal to the axial direction X of the rotary shaft **13**. The first board portion **51a** extends across the entirety of the first accommodation space S21 in the first direction Y and across the entirety of the first accommodation space S21 in the second direction Z. The second board portion **51b** extends across the entirety of the second accommodation space S22 in the first direction Y and across the entirety of the second accommodation space S22 in the second direction Z. The two capacitors **54** and the magnetic component **53** are accommodated in the second accommodation space S22.

[0043] The drive elements **52a** of the inverter circuit **52** perform switching operations to drive the motor **15**. The drive elements **52a** of the inverter circuit **52** are mounted on the first mounting surface **511** of the first board portion **51a** of the circuit board **51**.

Magnetic Component

[0044] As shown in FIG. 4, the magnetic component **53** includes a choke coil **55** and a conductive ring **56**. Hereinafter, the choke coil **55** will be referred to as a magnetic main body **551**. The magnetic main body **551** includes a magnetic core **60** formed of a magnetic material, a first coil **61**, and a second coil **62**. The coils **61**, **62** are wound around the magnetic core **60**.

[0045] The magnetic core **60** has the shape of a stadium or an ellipse. The magnetic core **60** is therefore a component extending in a longitudinal direction. The magnetic core **60** formed of a ferromagnetic material is, for example, a ferrite core. The magnetic core **60** includes a first spool portion **601**, a second spool portion **602**, and two coupling portions **603**. Each of the first spool portion **601** and the second spool portion **602** has the shape of a rectangular parallelepiped. The first spool portion **601** and the second spool portion **602** extend parallel to each other. One of the coupling portions **603** couples one end of the first spool portion **601** in the longitudinal direction to one end of the second spool portion **602** in the longitudinal direction. The other coupling portion **603** couples the other end of the first spool portion **601** in the longitudinal direction to the other end of the second spool portion **602** in the longitudinal direction.

[0046] The magnetic core **60** has two core main surfaces **60a**. The two core main surfaces **60a** are end faces of the magnetic core **60** in the axial direction and opposite surfaces of the magnetic core **60** in the thickness direction. The magnetic core **60** has two core side surfaces **60b**. The core side surfaces **60b** are formed by the outer surface of the coupling portion **603**. Each of the first coil **61** and the second coil **62** includes a winding wound around the coupling portions **603**. Each core side surface **60b** is a portion of an end face of the magnetic core **60** in the longitudinal direction, and is located between the winding of the first coil **61** and the winding of the second coil **62**, which are wound around the coupling portions **603**. One of the two core side surfaces **60b** faces the circuit board **51**, which will be discussed below.

[0047] One of the two core side surfaces **60b** connects ends on the same side of the two core main surfaces **60a** to each other. The other core side surface **60b** connects the ends on the opposite side of the two core main surfaces **60a** to each other. The surface area of each core side surface **60b** is smaller than the surface area of each core main surface **60a**.

[0048] The first coil **61** is wound around the first spool portion **601** of the magnetic core **60**. The two ends of the first coil **61** are two first lead wires **61a**, which are drawn out from the magnetic core **60**. The second coil **62** is wound around the second spool portion **602** of the magnetic core **60**.

The two ends of the second coil **62** are two second lead wires **62a**, which are drawn out from the magnetic core **60**. Each of the two first lead wires **61a** and the two second lead wires **62a** is drawn out along the core main surface **60a** of one of the coupling portions **603**, and protrudes outward beyond the core side surface **60b** of that coupling portion **603**.

[0049] As shown in FIG. 2, in the magnetic main body **551**, the dimension in the thickness direction of the magnetic core **60** including the first coil **61** and the second coil **62** is defined as a first dimension M1. In the magnetic main body **551**, the dimension of the core main surface **60a** in the longitudinal direction is defined as a second dimension M2. The first dimension M1 is smaller than the second dimension M2. Accordingly, the magnetic main body **551** is a component extending in the longitudinal direction.

[0050] A virtual surface T is established along one of the core side surfaces **60b**. The virtual plane T is a plane including the core side surface **60b** and is orthogonal to the longitudinal direction of the magnetic core **60**. Each of the first lead wires **61a** and the second lead wires **62a** extends linearly from the virtual plane T. The first lead wires **61a** and the second lead wires **62a** extend from the virtual plane T so as to overlap with one of the core side surfaces **60b** in the axial direction X.

[0051] As indicated by the long-dash double-short-dash lines in FIG. 4, the conductive ring **56** has the shape of a loop. The conductive ring **56** is made of a conductive material. The conductive ring **56** is made of, for example, copper or aluminum. The conductive ring **56** includes two first plate portions **56a** and two second plate portions **56b**.

[0052] Each of the two first plate portions **56a** and the two second plate portions **56b** has the shape of a rectangular flat plate. The two first plate portions **56a** face each other with the magnetic core **60**, the first coil **61**, and the second coil **62** interposed therebetween. One of the two first plate portions **56a** connects the respective ends of the two second plate portions **56b** to each other. The other first plate portion **56a** connects the opposite ends of the two second plate portions **56b** to each other. The two second plate portions **56b** face each other. The direction in which the two first plate portions **56a** face each other and the direction in which the two second plate portions **56b** face each other are orthogonal to each other.

[0053] Part of the magnetic main body **551** is disposed inside the conductive ring **56**. The axial direction of the magnetic core **60** and the axial direction of the conductive ring **56** are orthogonal to each other. The axial direction of the magnetic core **60** agrees with the first direction Y. The axial direction of the conductive ring **56** agrees with the axial direction X. The first spool portion **601** and the second spool portion **602** of the magnetic core **60**, a portion of the first coil **61** that is wound around the first spool portion **601**, and a portion of the second coil **62** that is wound around the second spool portion **602** are located inside the conductive ring **56**. The two first plate portions **56a** of the conductive ring **56** are disposed so as to sandwich the magnetic main body **551** in the axial direction of the magnetic core **60**. The two second plate portions **56b** of the conductive ring **56** are disposed so as to sandwich the magnetic main body **551** in the direction in which the first coil **61** and the second coil **62** are arranged.

[0054] When a normal mode current flows through the first coil **61** the second coil **62**, magnetic flux leaks from the magnetic core **60**. Accordingly, induced current flows in the conductive ring **56** to generate magnetic flux resisting changes in the leakage magnetic flux leaking from the magnetic core **60**. The induced current flowing through the conductive ring **56** is converted into thermal energy, so that normal mode noise is reduced. The magnetic component **53** is therefore an electronic component that removes noise. The magnetic component **53** includes the magnetic main body **551**, which acts to remove noise.

[0055] As shown in FIG. 2, the magnetic component **53** is mounted on the first mounting surface **511** of the second board portion **51b** of the circuit board **51**. The magnetic main body **551** of the magnetic component **53** is disposed in the second accommodation space S22 so as to face the outer peripheral surface **24a** of the motor housing member **24**. Therefore, the inverter accommodating

chamber **S2** includes the second accommodation space **S22**, which is arranged such that the magnetic main body **551** faces the outer peripheral surface **24a** of the motor housing member **24** in the radial direction **R** of the rotary shaft **13**.

[0056] Each of the two first lead wires **61a** and the two second lead wires **62a** of the magnetic main body **551** is electrically connected to the circuit board **51**. Therefore, each of the two first lead wires **61a** and the two second lead wires **62a** is a connection portion that electrically connects the magnetic main body **551** to the circuit board **51**. A portion of each of the two first lead wires **61a** and the two second lead wires **62a** that extends outward beyond the core side surface **60b** included in the virtual plane **T** is connected to the circuit board **51** without being bent.

[0057] The magnetic component **53** is mounted on the second board portion **51b** of the circuit board **51** and is accommodated in the second accommodation space **S22**. In other words, the magnetic main body **551** of the magnetic component **53** is disposed inside the projecting wall **111**. The axial direction of the magnetic core **60** is orthogonal to the axial direction **X** and agrees with the first direction **Y**. The magnetic main body **551** is disposed inside the projecting wall **111** such that the thickness direction of the magnetic core **60** is orthogonal to the axial direction **X**. One of the two core main surfaces **60a** faces the defining surface **240**, which is the outer peripheral surface **24a** of the motor housing member **24**.

[0058] One of the two first plate portions **56a** of the conductive ring **56** is located between the motor housing member **24** and the magnetic main body **551**.

[0059] The two core side surfaces **60b** of the magnetic core **60** includes one core side surface **60b** that is included in the virtual plane **T**, from which the first lead wires **61a** and the second lead wires **62a** extend, and faces the first mounting surface **511** of the second board portion **51b** of the circuit board **51**.

Capacitors

[0060] As shown in FIGS. **3** and **5**, each capacitor **54**, which is an electronic component, includes a capacitor main body **57** and lead wires **58**. The capacitor main body **57** of each capacitor **54** is a main body that removes noise. The capacitor main body **57** includes two electrodes **541** and a case **542** that accommodate the two electrodes **541**. The capacitor main body **57** has the shape of a rectangular parallelepiped. The capacitor main body **57** includes two capacitor main surfaces **57a**. The capacitor main surfaces **57a** are the largest of the six outer surfaces of the capacitor main body **57**. A direction orthogonal to the two capacitor main surfaces **57a** is defined as a thickness direction of the capacitor main body **57**. The capacitor main body **57** includes two capacitor end faces **57b**. The two capacitor end faces **57b** are outer surfaces that have the smallest areas among the six outer surfaces of the capacitor main body **57** and are continuous with the opposite ends of the capacitor main surfaces **57a** in the longitudinal direction. The dimension **N1** of each capacitor end face **57b** in the thickness direction of the capacitor main body **57** is smaller than the dimension **N2** of each capacitor main surface **57a** in the longitudinal direction.

[0061] The multiple lead wires **58** extend straight from one of the two capacitor end faces **57b**. The two capacitor end faces **57b** of each capacitor main body **57** include one capacitor end face **57b** that is included in the virtual plane **T**, from which the lead wires **58** extend, and faces the first mounting surface **511** of the second board portion **51b** of the circuit board **51**.

[0062] The capacitors **54** are mounted on the second board portion **51b** of the circuit board **51**. Specifically, each capacitor **54** is mounted on the circuit board **51** such that the capacitor end face **57b** from which the lead wires **58** extend faces the first mounting surface **511** of the second board portion **51b**.

[0063] Each capacitor **54** is mounted on the circuit board **51** such that the longitudinal direction of the capacitor main surface **57a** is orthogonal to the first mounting surface **511** of the circuit board **51**. In other words, each capacitor **54** is mounted on the circuit board **51** such that the longitudinal direction of the capacitor main surface **57a** agrees with the axial direction **X** of the motor housing member **24**. The thickness direction of the capacitor main body **57** agrees with the first direction **Y**.

Therefore, the thickness direction of the capacitor main body 57 is orthogonal to the axial direction X of the motor housing member 24. Thus, each capacitor 54 is disposed inside the projecting wall 111 such that the thickness direction of the capacitor main body 57 is orthogonal to the axial direction X. One of the two capacitor main surfaces 57a faces the defining surface 240 of the outer peripheral surface 24a of the motor housing member 24. Therefore, each capacitor 54 is accommodated in the second accommodation space S22. Thus, the inverter accommodating chamber S2 includes the second accommodation space S22, which is arranged such that each capacitor main body 57 faces the outer peripheral surface 24a of the motor housing member 24 in the radial direction R of the rotary shaft 13.

[0064] As shown in FIG. 3, the two capacitors 54 are arranged at the opposite sides of the magnetic component 53 in the second direction Z. In other words, the magnetic component 53 is located between the two capacitors 54 in the second direction Z. Therefore, the two capacitors 54 and the magnetic component 53 are arranged alongside one another in the second direction Z inside the projecting wall 111. That is, the two capacitors 54 and the magnetic component 53 are arranged alongside one another in the second direction Z at the same position in the first direction Y.

[0065] As shown in FIGS. 2 and 5, the connector 17 and the wire 17a are disposed at positions farther from the motor housing member 24 than the two capacitors 54 and the magnetic component 53 in the first direction Y. In other words, the two capacitors 54 and the magnetic component 53 are disposed at positions closer to the motor housing member 24 than the connector 17 and the wire 17a in the first direction Y.

[0066] Potting material P, which is a heat transfer member, is provided between the outer peripheral surface 24a of the motor housing member 24 and the group of the magnetic main body 551, the conductive ring 56, and the capacitors 54. The potting material P may be provided between the closing wall 26 and the core side surface 60b of the magnetic main body 551, from which the first lead wires 61a or the second lead wires 62a are not drawn out, and between the closing wall 26 and the capacitor end face 57b of each capacitor 54, from which the lead wires 58 are not drawn out.

Operation of the Present Embodiment

[0067] Operation of the present embodiment will now be described.

[0068] In FIG. 5, a comparative example is indicated by long-dash double-short-dash lines. In the comparative example, the capacitors 54 are disposed such that the lead wires 58 are bent and one of the capacitor main surfaces 57a faces the first mounting surface 511 of the second board portion 51b. In this case, the projected area of each capacitor main body 57 onto the circuit board 51 in the axial direction X of the rotary shaft 13 is the same as the size of the capacitor main surface 57a.

The projected area of each capacitor main body 57 onto the outer peripheral surface 24a of the motor housing member 24 in the radial direction R of the rotary shaft 13 is the same as the size of the capacitor end face 57b. Therefore, the projected area of each capacitor main body 57 onto the circuit board 51 in the axial direction X is larger than the projected area of the capacitor main body 57 onto the outer peripheral surface 24a of the motor housing member 24 in the radial direction R.

[0069] As indicated by the solid lines in FIG. 5, in the present embodiment, the capacitors 54 are disposed such that the lead wires 58 extend straight without being bent, and one of the capacitor end faces 57b faces the first mounting surface 511 of the second board portion 51b. In this case, the projected area of each capacitor main body 57 onto the circuit board 51 in the axial direction X of the rotary shaft 13 is the same as the size of the capacitor end face 57b. The projected area of each capacitor main body 57 onto the outer peripheral surface 24a of the motor housing member 24 in the radial direction R of the rotary shaft 13 is the same as the size of the capacitor main surface 57a. Therefore, the projected area of each capacitor main body 57 onto the circuit board 51 in the axial direction X is smaller than the projected area of the capacitor main body 57 onto the outer peripheral surface 24a of the motor housing member 24 in the radial direction R.

[0070] The lead wires 58 are disposed so as to overlap with the capacitor end faces 57b in the axial direction X of the rotary shaft 13. Thus, the lead wires 58 extend such that the projected area of

each capacitor main body **57** onto the circuit board **51** in the axial direction **X** is smaller than the projected area of the capacitor main body **57** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction **R** of the rotary shaft **13**.

[0071] In the present embodiment, as indicated by the solid lines in FIG. 2, the first lead wires **61a** and the second lead wires **62a** of the magnetic component **53** extend straight without being bent. The magnetic main body **551** is disposed such that one of the core side surfaces **60b** of the magnetic core **60** faces the first mounting surface **511** of the second board portion **51b**. In this case, the projected area of the magnetic main body **551** onto the circuit board **51** in the axial direction **X** of the rotary shaft **13** is substantially the same as the size of the core side surface **60b**. The projected area of the magnetic main body **551** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction **R** of the rotary shaft **13** is substantially the same as the size of each core main surface **60a**. Therefore, the projected area of the magnetic main body **551** onto the circuit board **51** in the axial direction **X** is smaller than the projected area of the magnetic main body **551** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction **R**.

[0072] The first lead wires **61a** and the second lead wires **62a** are disposed so as to overlap with the core side surface **60b** in the axial direction **X** of the rotary shaft **13**. Thus, the first lead wires **61a** and the second lead wires **62a** extend such that the projected area of the magnetic main body **551** onto the circuit board **51** in the axial direction **X** is smaller than the projected area of the magnetic main body **551** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction **R** of the rotary shaft **13**.

[0073] The connector **17** and the wire **17a** are disposed at positions farther from the motor housing member **24** than the two capacitors **54** and the magnetic component **53** in the first direction **Y**. Specifically, the second board portion **51b** accommodated in the inverter accommodating chamber **S2** includes a region **W**, on which the connector **17** is disposed via the wire **17a**, which connects the connector **17** to the circuit board **51**. The region **W** is located in a portion outward of the capacitors **54** and the magnetic component **53** in the first direction **Y**. In the comparative example, if the area **W** is located in a portion of the second board portion **51b** that is outward of the capacitors **54** and the magnetic component **53** in the first direction **Y**, the connector **17** and the wire **17a** are located farther from the motor housing member **24** in the first direction **Y** than in the case of the embodiment, as indicated by the long-dash double-short-dash lines in FIG. 5. In contrast, in the present embodiment, the area on the second board portion **51b** of the circuit board **51** on which the capacitors **54** and the magnetic component **53** are located is smaller than that in the comparative example. As a result, even if the region **W** for disposing the connector **17** via the wire **17a**, which connects the connector **17** to the circuit board **51**, is provided on the second board portion **51b**, the dimension of the second board portion **51b** in the first direction **Y** is smaller than that in the comparative example. Consequently, the dimension in the first direction **Y** of the projecting wall **111**, which accommodates the second board portion **51b**, is also smaller than that in the comparative example.

Advantages of the Present Embodiment

[0074] The present embodiment has the following advantages. [0075] (1) The capacitors **54** are disposed in the second accommodation space **S22** in the inverter housing member **25** and are mounted on the circuit board **51** such that the projected area of each capacitor main body **57** onto the second board portion **51b** in the axial direction **X** is smaller than the projected area of the capacitor main body **57** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction **R**. In other words, the longitudinal direction of each capacitor main body **57** extends in the axial direction **X**. As a result, when the housing **11** is seen in the axial direction **X**, the configuration reduces the dimension in the radial direction **R** of the rotary shaft **13** of the portion of the inverter housing member **25** that defines the second accommodation space **S22**. Consequently, the inverter housing member **25** suppresses vibration of the portion that defines the

second accommodation space S22, thereby suppressing the generation of noise due to such vibration. [0076] (2) The magnetic component 53 is disposed in the second accommodation space S22 in the inverter housing member 25 and is mounted on the circuit board 51 such that the projected area of the magnetic main body 551 onto the second board portion 51b in the axial direction X is smaller than the projected area of the magnetic main body 551 onto the outer peripheral surface 24a of the motor housing member 24 in the radial direction R. In other words, the longitudinal direction of the magnetic main body 551 extends in the axial direction X. As a result, when the housing 11 is seen in the axial direction X, the configuration reduces the dimension in the radial direction R of the rotary shaft 13 of the portion of the inverter housing member 25 that defines the second accommodation space S22. Consequently, the inverter housing member 25 suppresses vibration of the portion that defines the second accommodation space S22, thereby suppressing the generation of noise due to such vibration. [0077] (3) The second accommodation space S22 for accommodating the two capacitors 54 and the magnetic component 53 is defined in the inverter housing member 25 of the housing 11. The extension 28, which defines the second accommodation space S22 for accommodating these three components, acts as a rib of the housing 11. This further suppresses the vibration of the portion of the inverter housing member 25 that defines the second accommodation space S22. [0078] (4) The dimension N2 in the longitudinal direction of each capacitor main surface 57a of the capacitor main body 57 of each capacitor 54 is larger than the dimension N1 in the thickness direction of each capacitor end face 57b of the capacitor main body 57. The second dimension M2 in the longitudinal direction of the magnetic main body 551 of the magnetic component 53 is larger than the first dimension M1 in the thickness direction of the magnetic core 60. Even though the magnetic component 53 and the capacitors 54, which have such shapes, are disposed in the second accommodation space S22 of the inverter housing member 25, the configuration reduces the size of the portion of the inverter housing member 25 that defines the second accommodation space S22. Therefore, the vibration of the inverter housing member 25 is further suppressed. This configuration thus suppresses the generation of noise caused by vibration of the inverter housing member 25. [0079] (5) The capacitors 54 are disposed such that the thickness direction of each capacitor main body 57 is orthogonal to the axial direction X. The magnetic component 53 is disposed such that the axial direction of the magnetic core 60 in the magnetic main body 551 is orthogonal to the axial direction X. The two capacitors 54 and the magnetic component 53 are arranged alongside one another in the second direction Z. Thus, compared to a configuration in which the two capacitors 54 and the magnetic component 53 are arranged alongside one another in the first direction Y (i.e., the radial direction R), the second board portion 51b for mounting the two capacitors 54 and the magnetic component 53 is made compact in the radial direction R. Therefore, when the housing 11 is seen in the axial direction X, the configuration reduces the dimension in the radial direction R of the rotary shaft 13 of the portion of the inverter housing member 25 that defines the second accommodation space S22. Consequently, the inverter housing member 25 suppresses vibration of the portion that defines the second accommodation space S22, thereby suppressing the generation of noise due to such vibration. [0080] (6) The capacitors 54 are disposed such that the thickness direction of each capacitor main body 57 is orthogonal to the axial direction X. Thus, each capacitor end face 57b faces the circuit board 51, and the lead wires 58 extending from the capacitor end face 57b extend straight. The magnetic component 53 is disposed such that the axial direction of the magnetic core 60 in the magnetic main body 551 is orthogonal to the axial direction X. The first lead wires 61a and the second lead wires 62a of the magnetic component 53 extend straight from the core side surface 60b toward the circuit board 51. Accordingly, the lead wires 58, the first lead wires 61a, and the second lead wires 62a are not bent to be connected to the circuit board 51. Therefore, as compared to a case in which the lead wires 58, the first lead wires 61a, and the second lead wires 62a are provided with bent sections, the number of points on which stresses are concentrated is reduced in the lead wires 58, the first lead wires 61a, and the second lead wires 62a. [0081] (7) The

lead wires **58** of each capacitor **54** extend such that the projected area of the capacitor main body **57** onto the circuit board **51** in the axial direction X is smaller than the projected area of the capacitor main body **57** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction R of the rotary shaft **13**. The first lead wires **61a** and the second lead wires **62a** of the magnetic component **53** extend such that the projected area of the magnetic main body **551** onto the circuit board **51** in the axial direction X is smaller than the projected area of the magnetic main body **551** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction R of the rotary shaft **13**. This reduces the area of a portion of the circuit board **51** on which the capacitors **54** and the magnetic main body **551** are disposed. Accordingly, the dimension of the portion of the inverter housing member **25** that defines the second accommodation space S22 is reduced in the radial direction R of the rotary shaft **13**. [0082] (8) The potting material P is provided between the capacitor main bodies **57** and the outer peripheral surface **24a** of the motor housing member **24**. The potting material P is also provided between the magnetic main body **551** and the outer peripheral surface **24a** of the motor housing member **24**. The potting material P increases the heat dissipation area of the capacitor main bodies **57** and the magnetic main body **551**. Therefore, the capacitor main bodies **57** and the magnetic main body **551** are cooled efficiently.

Modifications

[0083] The above-described embodiment may be modified as described below. The above-described embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0084] The structure of the housing **11** may be changed.

[0085] For example, in the above-described embodiment, the partition wall **23**, the motor housing member **24**, the inverter housing member **25**, and the extension **28** are integrally formed as the housing body **20**. However, these components do not necessarily need to be integrally formed. Specifically, the motor housing member **24** and the inverter housing member **25** may be formed separately, and the extension **28** may be formed integrally with the motor housing member **24**. In this case, when the motor housing member **24** and the inverter housing member **25** are joined to each other, the second accommodation space S22 may be defined by the motor housing member **24** and the inverter housing member **25**.

[0086] Only one of the electronic components, either the magnetic component **53** or one of the two capacitors **54**, may be accommodated in the second accommodation space S22.

[0087] Electronic components other than the magnetic component **53** and the two capacitors **54**, for example, the drive elements **52a**, which form the inverter circuit **52**, may be accommodated in the second accommodation space S22.

[0088] The magnetic component **53** of the inverter **16** does not necessarily need to include the conductive ring **56**.

[0089] Among the magnetic component **53** and the two capacitors **54**, the magnetic component **53** and one of the capacitors **54** may be arranged alongside each other in the second direction Z, and the other capacitor **54** may be arranged on the outer side in the first direction Y of the capacitor **54** and the magnetic component **53** arranged alongside each other in the second direction Z.

[0090] Among the magnetic component **53** and the two capacitors **54**, the two capacitors **54** may be arranged alongside each other in the second direction Z, and the magnetic component **53** may be arranged on the outer side in the first direction Y of the two capacitors **54** arranged alongside each other in the second direction Z.

[0091] The magnetic component **53** and the two capacitors **54** may all be arranged alongside one another in the first direction Y.

[0092] The lead wires **58** may include bent portions as long as the projected area of each capacitor main body **57** onto the second board portion **51b** in the axial direction X is smaller than the projected area of the capacitor main body **57** onto the outer peripheral surface **24a** of the motor

housing member **24** in the radial direction R of the rotary shaft **13**. The first lead wires **61a** and the second lead wires **62a** may have bent portions as long as the projected area of the magnetic main body **551** onto the second board portion **51b** in the axial direction X is smaller than the projected area of the magnetic main body **551** onto the outer peripheral surface **24a** of the motor housing member **24** in the radial direction R of the rotary shaft **13**.

[0093] The heat transfer member may be a thermal grease or a thermal sheet, rather than the potting material P.

[0094] The motor-driven compressor **10** may also be used for purposes other than a vehicle air conditioner. For example, the motor-driven compressor **10** may be mounted on a fuel cell electric vehicle. The motor-driven compressor **10** uses the compression unit **14** to compress air, which is fluid, supplied to the fuel cell.

[0095] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A motor-driven compressor, comprising: a rotary shaft; a motor configured to rotate the rotary shaft; a compression unit configured to compress a fluid by rotation of the rotary shaft; an inverter configured to drive the motor; and a housing that accommodates the rotary shaft, the compression unit, the motor, and the inverter, wherein the inverter includes a circuit board on which a drive element configured to drive the motor and an electronic component configured to remove noise are mounted, the housing includes: a motor housing member that accommodates the motor and defines a suction chamber configured to draw in the fluid; a compression unit housing member that accommodates the compression unit and is configured to discharge the compressed fluid; an inverter housing member that defines an inverter accommodating chamber that accommodates the inverter; and a partition wall that separates the suction chamber and the inverter accommodating chamber from each other, the electronic component includes: a main body that acts to remove noise; and a connection portion that electrically connects the main body and the circuit board to each other, the inverter accommodation chamber includes: a first accommodation space in which the circuit board is disposed to face the partition wall; and a second accommodation space that opens in an axial direction of the rotary shaft and is arranged such that the main body faces an outer peripheral surface of the motor housing member in a radial direction of the rotary shaft, and a projected area of the main body onto the circuit board in the axial direction is smaller than a projected area of the main body onto the outer peripheral surface of the motor housing member in the radial direction.

2. The motor-driven compressor according to claim 1, wherein the electronic component includes a capacitor, the capacitor is accommodated in the second accommodation space, the main body is a capacitor main body of the capacitor, the capacitor main body including two electrodes and a case accommodating the two electrodes, and a projected area of the capacitor main body onto the circuit board in the axial direction is smaller than a projected area of the capacitor main body onto the outer peripheral surface of the motor housing member in the radial direction of the rotary shaft.

3. The motor-driven compressor according to claim 1, wherein the electronic component includes a magnetic component, the magnetic component is accommodated in the second accommodation

space, the main body is a magnetic main body of the magnetic component, the magnetic main body including a magnetic core formed of a magnetic material and a coil wound around the magnetic core, and a projected area of the magnetic main body onto the circuit board in the axial direction is smaller than a projected area of the magnetic main body onto the outer peripheral surface of the motor housing member in the radial direction of the rotary shaft.

4. The motor-driven compressor according to claim 1, wherein the connection portion is a lead wire, and the lead wire extends such that a projected area of the main body onto the circuit board in the axial direction is smaller than a projected area of the main body onto the outer peripheral surface of the motor housing member in the radial direction of the rotary shaft.

5. The motor-driven compressor according to claim 1, wherein a heat transfer member is provided between the main body and the outer peripheral surface of the motor housing member.
